Fourier Analysis.

Definition: If f is integrable on $[a,b]$. $L = b-a$.

Def

n^{th} Fourier coefficient of f:

$$\hat{f}(n) = \frac{1}{L}\int_a^b f(x) e^{-\frac{2\pi i n x}{L}} dx \qquad n \in \mathbb{Z}$$

Fourier series of f: $\sum_{n=-\infty}^{\infty} \hat{f}(n)\, e^{2\pi i n x/L}$.

Some notations:

$$a_n = \hat{f}(n). \quad ; \quad f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n x/L}.$$

Example 1: $f(\theta) = \theta$. $-\pi \le \theta \le \pi$.

$$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \theta\, e^{-in\theta} d\theta = \frac{(-1)^{n+1}}{in}$$

$$f(\theta) \sim \sum_{n\neq 0} \frac{(-1)^{n+1}}{in} e^{in\theta} = \sum_{n\neq 0} \frac{(-1)^{n+1}}{in}(\cos n\theta + i \sin n\theta)$$

$$= 2\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{\sin n\theta}{n}.$$

Example 2: $f(\theta) = \frac{(\pi-\theta)^2}{4}$.

Caculate: $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{(\pi-\theta)^2}{4} e^{-in\theta} d\theta$.

Then we have: $f(\theta) \sim \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos n\theta}{n^2}$.

Example 4: Dirichlet kernel: $D_N(x) = \sum_{n=-N}^{N} e^{inx}$ $\qquad x \in [-\pi, \pi]$.

Caculate and we have: $D_N(x) = \frac{\sin(N+\frac{1}{2})x}{\sin(\frac{x}{2})}$.

Also: $\begin{cases} a_n = 1. & |n| \le N. \\ a_n = 0. & |n| > N \end{cases}$

Example 5: Poisson kernel: $\theta \in [-\pi, \pi]$. $0 \le r < 1$.

$$P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta}.$$

Caculate: $P_r(\theta) = \sum_{n=0}^{\infty} w^n + \sum_{n=1}^{\infty} \bar{w}^n$. $\qquad w = re^{i\theta}$.

$$= \frac{1}{1-w} + \frac{\bar{w}}{1-\bar{w}} = \frac{1-|w|^2}{|1-w|^2} = \frac{1-r^2}{1-2r\cos\theta + r^2}.$$

Also: $\hat{P}_r(n) = r^{|n|}$

Theorem 2.1

f is integrable, $\hat{f}(n)=0 \quad \forall n \in \mathbb{Z}$.

if f is continuous at θ_0, then $f(\theta_0)=0$.

pf: contradiction : When f is real-valued.

Assume that f is defined on $[-\pi,\pi]$. , $\theta_0=0$. And $f(\theta_0)>0$.

choose $0<\delta<\frac{\pi}{2}$. so that $f(\theta)>\frac{f(0)}{2}$ when $|\theta|<\delta$.

(1) Define $p(\theta)=\varepsilon+\cos\theta$. where $\varepsilon>0$. that $|p(\theta)|<1-\frac{\varepsilon}{2}$. $\forall \quad \delta\le|\theta|\le\pi$.

(2) Then choose $\eta>0$. s.t. $\eta<\delta$ and $p(\theta)\ge 1+\frac{\varepsilon}{2}$ when $|\theta|<\eta$.

(3) select B. s.t. $|f(\theta)|\le B$. Def $p_k(\theta)=[p(\theta)]^k$

Since $\hat{f}(n)=0$. $\forall n\in\mathbb{Z}$. we have $\int_{-\pi}^{\pi} f(\theta)p_k(\theta)d\theta=0$. $\forall k\in\mathbb{N}_+$.

In fact: $\hat{f}(k)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}dx=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)[\cos kx - i\sin kx]dx=0$

$$\int_{-\pi}^{\pi} f(\theta)p_k(\theta)d\theta=\int_{-\pi}^{\pi} f(\theta)(\varepsilon+\cos\theta)^k d\theta=0$$

(In fact: $\int_{-\pi}^{\pi} f(x)\cos kx\,dx=0$. ; $\int_{-\pi}^{\pi} f(x)\sin kx\,dx=0$.

Then $\int_{-\pi}^{\pi} f(\theta)(\varepsilon+\cos\theta)^k d\theta=\int_{-\pi}^{\pi} f(\theta)\left(\varepsilon+\frac{e^{i\theta}+e^{-i\theta}}{2}\right)^k d\theta=$ "some linear sum of $A_t\int_{-\pi}^{\pi} f(\theta)\varepsilon^{n_t}e^{ik\theta}$" $=0$)

However:

$$\begin{cases} \left|\int_{\delta\le|\theta|} f(\theta)p_k(\theta)d\theta\right| \le 2\pi B(1-\frac{\varepsilon}{2})^k, \\ \int_{\eta\le|\theta|<\delta} f(\theta)p_k(\theta)d\theta \ge 0. \\ \int_{|\theta|<\eta} f(\theta)p_k(\theta)d\theta \ge 2\eta\cdot\frac{f(0)}{2}(1+\frac{\varepsilon}{2})^k. \end{cases}$$

$\Rightarrow \int_{-\pi}^{\pi} f(\theta)p_k(\theta)d\theta\to\infty$ when $k\to\infty$. It leads to paradox!

When f isn't real-valued.

Let $f(\theta)=u(\theta)+iv(\theta)$. u, v is real-valued.

then $u(\theta)=\frac{1}{2}(f(\theta)+\overline{f}(\theta))$ $v(\theta)=\frac{1}{2i}(f(\theta)-\overline{f}(\theta))$.

Since $\hat{f}(n)=\overline{\hat{f}(n)}$. we have $\hat{f}(n)=0$. $\Rightarrow$ $\hat{u}(n)=\hat{v}(n)=0$

so $u(\theta_0)=0$. $v(\theta_0)=0$ when f is continuous on θ_0

Corrollary. f is continuous on the circle and $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| < \infty$.

Then we have: $\lim_{N\to\infty} S_N(f)(\theta) = f(\theta)$ uniformly in θ

Pf: $S_N(f)(\theta) = \frac{1}{2\pi}\sum_{n=-N}^{N} \left(\int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta\right) \cdot e^{in\theta}$.

Since: $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| < \infty$. and $\forall \theta$, $|\hat{f}(n)e^{in\theta}| = |\hat{f}(n)|$

we have $g(\theta) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{in\theta}$ is well-defined and continuous on the circle. (Weierstrass M-Thm).

And. the fourier coefficients of g:

$$\hat{g}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(\theta) e^{-in\theta} d\theta$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{i(k-n)\theta} d\theta = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} \hat{f}(k) \int_{-\pi}^{\pi} e^{i(k-n)\theta} d\theta$$

$$= \frac{1}{2\pi}\hat{f}(n)\int_{-\pi}^{\pi} 1\, d\theta = \hat{f}(n).$$

(Notice that we can interchange the sum with the integral for the uniform convergence).

Def: $h(\theta) = f(\theta) - g(\theta)$ $\theta \in [-\pi, \pi]$. obviously h is continuous and $\hat{h}(n) = 0$ $\forall n \in \mathbb{Z}$

So. $h = 0$. And that is $f(\theta) = g(\theta) = \lim_{N\to\infty} S_N(f)(\theta)$ □.

P.S. The proof of Weierstrass - M condition

1. $\forall n \in \mathbb{N}. \quad \forall x \in \mathbb{R}. \quad |f_n(x)| \le M_n.$
2. $\sum_{n=1}^{\infty} M_n < \infty$

Then, we have. $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly.

pf: Let $S_n(x) = \sum_{k=1}^{n} f_k(x)$, Then: $S(x) \triangleq \lim_{n\to\infty} S_n(x)$ (for a fixed x, $\{S_n(x)\}_n$ is a Cauchy sequence)

our goal is $\sup_{x\in\mathbb{R}} |S_n(x) - S(x)| \to 0 \quad (n \to \infty)$

$\forall \varepsilon > 0., \exists N > 0., \forall m > n \ge N.$ then: $\sum_{k=n+1}^{m} M_k < \varepsilon.$

That is: $|S_m(x) - S_n(x)| = \left|\sum_{k=n+1}^{m} f_k(x)\right| \le \sum_{k=n+1}^{m} M_k < \varepsilon., \quad \forall x \in \mathbb{R}.$

Let $m \to \infty.$ we have $|S(x) - S_n(x)| < \varepsilon \quad \forall x \in \mathbb{R}.$

So $\sum_{n=1}^{\infty} f_n(x) = S(x)$ uniformly. □.

Then we find out the condition of $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| < \infty$.

Corollary

Corollary 2.4. If: $f \in C^2(\mathbb{R}/2\pi\mathbb{Z})$, we have $\hat{f}(n) = O(\frac{1}{|n|^2})$ $n \to \infty$.

pf: $2\pi \hat{f}(n) = \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = \int_0^{2\pi} f(\theta)\, d\left(\frac{e^{-in\theta}}{-in}\right)$

$= \frac{1}{in} \int_0^{2\pi} f'(\theta) e^{-in\theta} d\theta = \frac{1}{in} \int_0^{2\pi} f'(\theta) \cdot d\left(\frac{e^{-in\theta}}{-in}\right)$.

$= \frac{-1}{n^2} \int_0^{2\pi} f''(\theta) e^{-in\theta} d\theta$.

So $2\pi |n^2| |\hat{f}(n)| = \left| \int_0^{2\pi} f''(\theta) e^{-in\theta} d\theta \right| \leq \int_0^{2\pi} |f''(\theta)| d\theta \leq C$

So $\hat{f}(n) = O(\frac{1}{|n|^2})$ $(n \to \infty)$. □.

p.s. We have $\hat{f}'(n) = in \hat{f}(n)$ $\forall n \in \mathbb{Z}$.

1). If $f \in C^1$. Then $\lim_{N\to\infty} S_N(f)(\theta) = f(\theta)$ uniformly. (Use Cauchy inequation { Bessel inequation.

2) If f satisfies: for a fixed α., $\alpha > \frac{1}{2}$.

$\sup_\theta |f(\theta+t) - f(\theta)| \leq A|t|^\alpha$. $\forall t \in \mathbb{R}$.

Then $\lim_{N\to\infty} S_N(f)(\theta) = f(\theta)$.

Convolutions:

Def: if f and g are 2π-periodic. on $\mathbb{R}$.

$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x-y) dy = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) g(y) dy$.

So we have following findings

$S_N(f)(x) = \sum_{n=-N}^{N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \right) e^{inx}$

$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \sum_{n=-N}^{N} e^{in(x-y)} dy$.

$= (f * D_N)(x)$.

$D_N(x) := \sum_{n=-N}^{N} e^{inx}$ is called Nth. Dirchlet Kernel.

Proposition: (Important). f, g, h are 2π-periodic & Riemann intergrable.

(i). $f * (g+h) = (f*g) + (f*h)$

(ii) $(cf) * g = c(f*g) = f*(cg) \quad \forall c \in \mathbb{C}$

(iii) $f * g = g * f$

(iv). $(f*g)*h = f*(g*h)$.

(v). $(f*g)$ is continuous.

(vi) $\widehat{f*g}(n) = \hat{f}(n)\,\hat{g}(n)$

(i). $f*(g+h) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(y)\,g(x-y) + f(y)\,h(x-y))\,dy = f*g + f*h.$

(ii) trival.

(iii). $f*g = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,g(x-y)\,dy$

$g*f = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(y)\,f(x-y)\,dy. \xrightarrow[y=x-z]{z=x-y} \frac{1}{2\pi}\int_{x+\pi}^{x-\pi} g(x-z)f(z)\,dz.$

$$= \frac{1}{2\pi}\int_{x-\pi}^{x+\pi} f(z)\,g(x-z)\,dz$$

$$= \frac{1}{2\pi}\left(\int_{x-\pi}^{-\pi} + \int_{-\pi}^{\pi} + \int_{\pi}^{x+\pi}\right)\left(f(z)\,g(x-z)\,dz\right)$$

$$\int_{x-\pi}^{-\pi} f(z)g(x-z)\,dz \overset{t=z+2\pi}{=\!=\!=} \int_{x+\pi}^{\pi} f(t)\,g(x-t)\,dt = -\int_{x+\pi}^{\pi} f(z)\,g(x-z)\,dz \Rightarrow g*f = f*g.$$

(iv.) $(f*g)*h = \frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi(y)\,h(x-y)\,dy.$

(v.) ~~$f*g = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\,g(x-y)\,dy. = \varphi(x)$~~

~~$\varphi(x+h) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\left[g(x+h-y)\right]dy.$~~

~~$\overset{z=y-h}{=\!=\!=} \frac{1}{2\pi}\int_{-\pi-h}^{\pi-h} f(z+h)\,g(x-z)\,dz$~~

Good kernels:

Def: $\{K_n(x)\}_{n\geq 1}$ on the circle, they are good kernels if

(a) For all $n \geq 1$ $\quad \frac{1}{2\pi}\int_{-\pi}^{\pi} K_n(x)\,dx = 1$

(b) $\exists M > 0$ s.t. $\forall n \geq 1$. $\quad \int_{-\pi}^{\pi} |K_n(x)|\,dx \leq M$.

(c) $\forall \delta > 0$ $\quad \int_{\delta \leq |x| \leq \pi} |K_n(x)| \to 0$ as $n \to \infty$

Thm 4.1. $\{K_n\}_{n\geq 1}$ is a family of good kernels.

f is intergrable on the circle then. $\lim_{n\to\infty} (f * K_n)(x) = f(x)$

if f is con at x.

Pf: $\forall \varepsilon > 0$. choose $\delta > 0$. so that $|y| < \delta \Rightarrow |f(x-y) - f(x)| < \varepsilon$. note that $|f(x)| \leq B$ for all x

$$|(f * K_n)(x) - f(x)| \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |K_n(y)| \left|f(x-y) - f(x)\right| dy.$$

$$\leq \frac{1}{2\pi}\int_{|y|<\delta} |K_n(y)|\,\varepsilon\,dy + \frac{1}{2\pi}\int_{\delta \leq |x| \leq \pi} |K_n(y)|\,2B\,dy$$

$\int_{|y|<\delta} |K_n(y)| < M$; $\quad \int_{\delta \leq |x| \leq \pi} |K_n(y)| < \varepsilon$. $(n \to \infty)$

$< C\varepsilon$. $\quad C$ is an independant constance

So: $\lim_{n\to\infty} (f * K_n)(x) = f(x)$

If f is continuous on the circle then it is uniformly continuous.

So δ is chosen independently $\Rightarrow f * K_n \to f$ uniformly.

Two other summations

1. Cesàro sum: let $\{c_n\}_n \subseteq \mathbb{C}$, $S_n = \sum_{k=0}^{n} c_k$.

Def: $\sigma_N = \frac{1}{N}\sum_{k=0}^{N-1} S_k$

If $\lim_{n\to\infty} \sigma_n = \sigma$. then we call $\sum c_n$ is Cesàro summable to σ.

And from that we have

Fejér's Thm: ① Fejér kernel: $F_N(x) = \dfrac{D_0(x) + \ldots + D_{N-1}(x)}{N}$

where $D_N(x)$ is Dirchlet kernel.

p.s. We wanted: $\lim_{N\to\infty} \sum_{n=-N}^{N} \hat{f}(n)\, e^{in\theta} \to f(\theta)$ (it failed at times)

So we use: $\sigma_N(f)(x) = \frac{1}{N}[S_0(f)(x) + \ldots + S_{N-1}(f)(x)]$

Since $S_N(f)(x) = f * D_n(x)$

we have $\sigma_N(f) = f * F_n$.

Lemma. $F_N(x) = \frac{1}{N} \frac{\sin^2\left(\frac{Nx}{2}\right)}{\sin^2\left(\frac{x}{2}\right)}$ is a good kernel.

Pf: $F_N(x) = \frac{1}{N}\left(D_0(x) + \dots + D_{N-1}(x)\right)$, Since $D_k(x) = \sum_{j=-k}^{k} e^{ijx} = \sum_{j=-k}^{k} \cos jx + i(\sin jx)$

$$= \sum_{j=-k}^{k} \cos jx \cdot$$

$$\cos jx = \frac{1}{2\sin\frac{x}{2}}\left[\sin(k+\tfrac{1}{2})x - \sin(k-\tfrac{1}{2})x\right] \Rightarrow D_k(x) = \frac{1}{\sin\frac{x}{2}} \sin(k+\tfrac{1}{2})x.$$

$$\frac{1}{N}\sum_{k=0}^{N-1} D_k(x) = \frac{1}{\sin\frac{x}{2} N}\sum_{k=0}^{N-1} \sin(k+\tfrac{1}{2})x \cdot = \frac{1}{\sin\frac{x}{2}} \cdot \frac{1}{-2\sin\frac{x}{2}}\left[\cos(N+\cdot)x - 1\right]$$

$$= \frac{\sin^2\left(\frac{Nx}{2}\right)}{N\sin^2\left(\frac{x}{2}\right)}$$

1) $\frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(x)\,dx = 1$ $\left(\frac{1}{2\pi}\int_{-\pi}^{\pi} D_k(x) = 1\right)$

2) $\int_{\delta \le |x| \le \pi} |F_N(x)|\,dx \le \int_{\delta \le |x| \le \pi} \frac{1}{N C_\delta''} \to 0$ $(N \to \infty)$

With the property of good kernels. we have

Thm: 1). If f is integrable on the circle, the $\sigma_N(f)(x) = (f * F_n)(x) \to f(x)$ when f is con at x.

2) If f is con on the circle, $\sigma_N(f) \to f$ uniformly.

3). If. $\hat{f}(n) = 0$ $\forall n \in \mathbb{Z}$. (then $\sigma_N(f) = 0$.) So $f = 0$ at continuous points.

4). If f is continuous on the circle, then $\forall \varepsilon > 0$. here exists $P(x)$, such that $|f(x) - P(x)| < \varepsilon$ for all $x \in [-\pi, \pi]$. $(P(x) = \sigma_N(f)(x))$

it is 三角多项式...

2. Abel means

Def: $\sum_{k=0}^{\infty} c_k$ is Abel summable to s: if $\forall\, r\in[0,1)$

$A(r)=\sum_{k=0}^{\infty} c_k r^k$ converges. and $\lim_{r\to 1} A(r)=s$

As same as 1. we consider: $A_r(f)(\theta)=\sum_{n=-\infty}^{\infty} r^{|n|} a_n e^{in\theta}$ instead of $\sum_{n=-\infty}^{\infty} a_n e^{in\theta}$
$(0\le r<1)$

$a_n=\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx$

If f is integrable. then $|a_n|$ is uniformly bounded in n (有界)

So $A_r(f)$ converges absolutely. and uniformly.

Also, $A_r(f)(\theta)=(f * P_r)(\theta)$. (Proof is obvious).

$P_r(\theta)$ is called Poisson kernel. $\triangleq \sum_{n=-\infty}^{\infty} r^{|n|}e^{in\theta}$

$P_r(\theta)=\sum_{n=0}^{\infty}\omega^n+\sum_{n=1}^{\infty}\bar{\omega}^n \quad (\omega=re^{i\theta}) = \frac{1-r^2}{1-2r\cos\theta+r^2}$.

$\forall\, r\in[0,1)$. we have. 1). $\frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta)d\theta=\frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_{n=-\infty}^{\infty} r^{|n|}e^{in\theta}\,d\theta$

$=\frac{1}{2\pi}\sum_{n=-\infty}^{\infty}\int_{-\pi}^{\pi} r^{|n|}e^{in\theta}d\theta = 1$

2) $P_r(\theta)\ge 0$

3) $\forall\varepsilon>0, \int_{\delta<|\theta|\le\pi} P_r(x)\,dx = \int_{\delta<|\theta|\le\pi}\frac{1-r^2}{(1-r)^2+2r(1-\cos\theta)}d\theta$. Hence if $\frac{1}{2}\le r<1$

$\le \int_{\delta<|\theta|\le\pi}\frac{1-r^2}{C_\delta}d\theta<\varepsilon$. $\delta\le|\theta|\le\pi$.